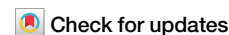




Organic photodiode receiver for a 3.6 gigabit per second visible light communication link



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Organic semiconductors offer a compelling platform for high-volume optoelectronic technologies with mechanical flexibility, tuneable properties, and potential for low-cost, sustainable production. Organic photodiodes are promising receivers for visible light communication systems due to their tuneable optical and electronic properties and inexpensive integration. However, their use in high-speed data transmission has been limited by material constraints, including low charge-carrier mobility and parasitic capacitance in thin-film structures. Here we show high-speed organic photodiodes using targeted device engineering that addresses these bandwidth-limiting factors. The devices achieve a record visible light communication data rate of 3.6 gigabits per second over a one-meter link using a single organic photodiode, more than twenty times faster than the previous best visible light communication data rate with organic receivers. These findings establish a new performance level for organic receivers and open a pathway towards their integration in high-speed, flexible, and sustainable applications for the Internet of Things and optical wireless sensors.

The rapid growth of smart devices has led to a huge demand for wireless communications, placing high and potentially unsustainable demand on existing radio frequency (RF) technologies. Visible light communication (VLC) is a novel technology that can complement current RF systems^{1,2}. VLC utilizes an optical spectrum that is approximately 2600 times larger than the entire RF spectrum³, and is not licenced (i.e. freely available). Additionally, VLC is less susceptible to electromagnetic interference making it more secure for data communications. VLC decreases the necessity for using additional power sources because it utilizes visible light which is available in many environments for lighting showing its potential to be multifunctional and energy efficient⁴. Moreover, VLC can be integrated with novel transmitters and receivers such as organic light-emitting diodes and organic solar cells to provide energy-efficient and sustainable applications^{5,6}.

Organic photodetectors (OPDs) are promising receivers for VLC due to their low-cost fabrication, mechanical flexibility^{7–9}, and tuneable optical and electronic properties through molecular design^{10–13}. They also offer the potential for seamless integration into everyday objects¹⁴. Organic devices represent promising components for industrial applications, image sensor-based applications, vehicle to vehicle and indoor optical communications^{15–18}. Furthermore, wearable healthcare technologies, where lightweight, flexible, and conformable devices can accurately monitor

physiological signals on irregular or curved surfaces or even when integrated within the human body¹⁹.

Organic optoelectronic devices have generally been regarded as slow devices, especially compared to their inorganic counterparts, but progress has been made to enhance their response speeds²⁰. In 2015, Zhang et al. achieved a receiver bandwidth of 1.3 MHz and a data rate of 42.3 Mbps with organic solar cells based on PTB7:PC₇₁BM material blends as VLC receivers while simultaneously harvesting energy from transmitted light²¹. More recently Tavakkolnia et al. expanded the work and achieved a bandwidth of 2.77 MHz with a data rate of 147.5 Mbps for a single organic solar cell and 363 Mbps with four organic solar cells by multiple-input multiple output (MIMO)⁶. In 2023, Zhu et al. demonstrated a bandwidth of 4 MHz with a data rate of 80 Mbps by making 1 mm² OPDs using novel non-fullerene acceptor¹⁸. In 2024, Saggar et al. improved the bandwidth of BHJ type OPDs to 4.5 MHz²². OPD bandwidths are potentially much higher in very small devices (e.g. Peumans et al. reported OPDs with a cut-off frequency of (430 ± 40) MHz by making a device of only 0.1-mm diameter and a complicated structure of 64 layers²³) but such a small area device would limit the signal to noise of a wireless VLC receiver. So, despite recent progress, the data rates of organic receivers remain lower than those of their inorganic counterparts, which are commercially preferred due to their superior speed. However, inorganic semiconductor-based optoelectronic detectors (e.g.,

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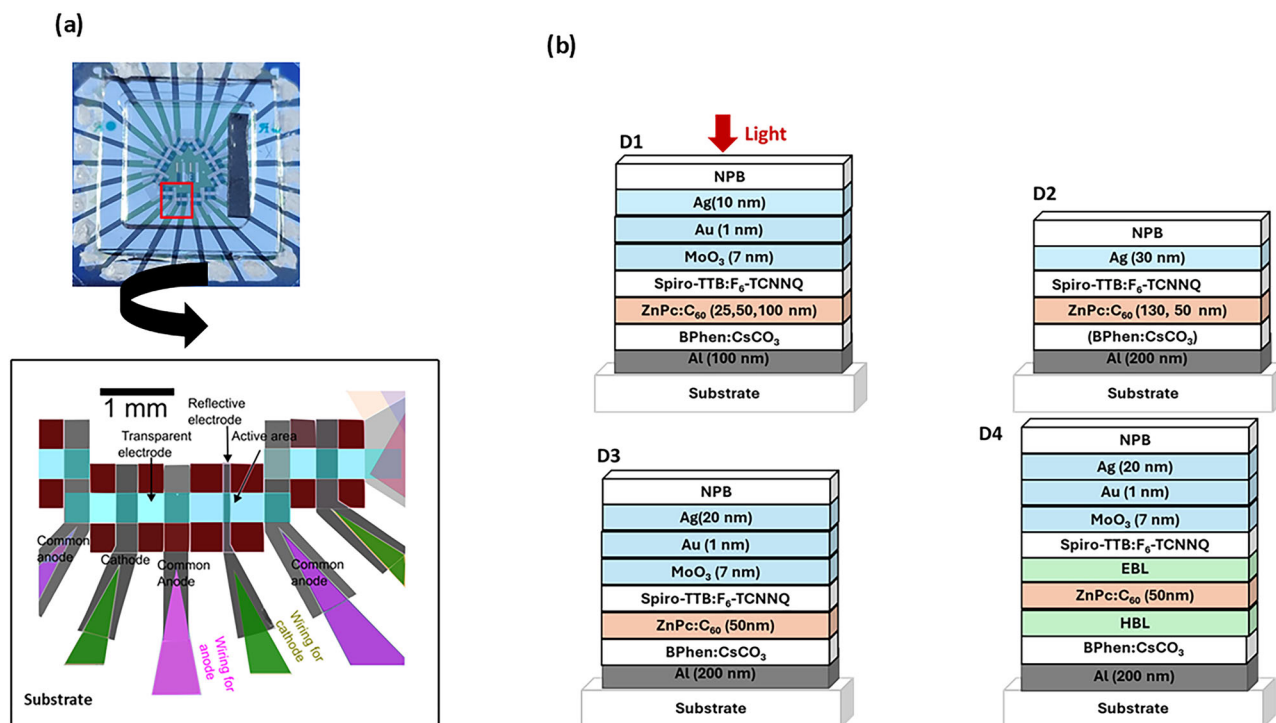


Fig. 1 | Overview of OPDs investigated in this work. a Photograph of device and schematic of the top view. **b** Different structures of OPDs explored in this work.

photodiodes and solar cells) require high-energy manufacturing processes and in some cases rely on toxic materials^{24,25}.

We report here a major advance in data rates of communication with OPDs by addressing key factors limiting their speed. It has been observed that the response speed of an OPD is limited by its electrical time constant, $\tau = RC$, where C is the device capacitance and R is the total circuit resistance²⁶. Reducing τ leads to a faster speed which can be achieved by decreasing the device area and increasing the active layer thickness to reduce the geometric capacitance.

However, the former reduces the amount of light detected, and the latter increases the transit time, leading to trade-offs in performance. The transit time of charges in the device, is limited by the low mobility of organic semiconductors^{27,28} and the thickness of the device. Applying a reverse bias leads to faster charge extraction and hence a faster response speed. Here, we explore the effect of active area thicknesses and size, electrode design, and applied electric field. We use these insights to develop OPDs with high modulation bandwidth and use them to demonstrate a record data rate of 3.6 Gbps over a 1 m link and 3.3 Gbps over a 3 m link for a single OPD receiver.

Results and Discussion

Overview of different structures of investigated OPDs

Figure 1 and Table 1 summarize the devices investigated in this study. Four different structures of OPDs were investigated in this work to explore the effect of active layer thickness, electrode design and the incorporation of electron and hole blocking layers on device performance.

All devices are in top illuminating configuration utilized an optical cavity structure with a reflective bottom electrode and a semi-transparent silver-based top electrode due to silver having a higher conductivity than indium-tin oxide (ITO)²⁹. All OPDs were designed using a doped transport layers based on an active area of zinc-phthalocyanine (ZnPc) as an electron donor and fullerene (C₆₀) as an electron acceptor²⁷. In addition, devices of two different sizes, 0.13 mm² and 0.05 mm², were fabricated to explore the effect of geometric capacitance on the bandwidth and data rate.

Thermal management

As discussed in the introduction section, applying a reverse bias voltage can be used to reduce transit time and therefore decrease the response time of the OPD. Supplementary Fig. 1 shows dark current as a function of reverse voltage for devices fabricated on glass and silicon substrates with active layer thicknesses of 25 nm, 50 nm, 100 nm and 130 nm. Devices on a glass substrate using 25 nm and 50 nm active layers showed degradation at -9 V and -13 V respectively as indicated by fluctuations in the curves (Supplementary Fig. 1a). However, this degradation was not observed for devices using thicker active layers of 100 nm and 130 nm. The applied electric field in devices using the thinner active layer thicknesses is stronger and can reach higher current densities by increasing the voltage, resulting in heat generation and increased degradation. To manage the thermal effect for devices using thinner active layers silicon substrates with higher thermal conductivity³⁰ was therefore chosen. A high reverse bias was used to generate a strong electric field, which reduces the carrier transit time and so improves the response speed and bandwidth of the OPD. High reverse bias can cause charge injection and device heating, so we make our devices on substrates with high thermal conductivity (silicon), to help dissipate heat. In addition, to dissipate Joule heating at high reverse voltages, the OPDs were mounted on a custom-made holder equipped with a heat sink and a fan.

Effect of active layer thickness: Trade-off between time constant, τ , and charge transit time

First, to explore the effect of active layer thickness on device performance the device structure D1 with three different thicknesses of 25 nm, 50 nm and 100 nm (see Fig. 1b) was selected. Because thin silver tends to form islands, we introduced MoO₃ (7 nm) and Au (1 nm) wetting layers, as suggested in ref. 29 to improve film continuity and electrical conductivity.

We initially set the OPD active area to 0.05 mm². To assess the frequency response of OPDs a modulated fast laser with a wavelength of 635 nm was used to illuminate the OPDs. Their bandwidth was defined to be the frequency at which the power gain decreases 6 dB from the value at low frequency (Supplementary Fig. 2) (more details of the measurements are provided in the Method section). Figure 2a shows the bandwidth as a function of voltage for devices with active layer thicknesses of 25 nm, 50 nm

Table 1 | Summary of different OPD structures

Device type	Active layer thickness (nm)	Semi-transparent electrode	Bottom electrode thickness (nm)	Blocking layers
D1	25, 50, 100	MoO ₃ -Au-Ag (10 nm)	Al (100)	None
D2	130, 50	Ag (30 nm)	Al (200)	None
D3	50	MoO ₃ -Au-Ag (20 nm)	Al (200)	None
D4	50	MoO ₃ -Au-Ag (20 nm)	Al (200)	BPhen-NPB

and 100 nm. We find the bandwidth improves by increasing the electric field for all devices. At -19 V the -6 dB bandwidth measured to be 32 MHz for the 100 nm device, 79 MHz for the 50 nm device and 71 MHz for the 25 nm device. The bandwidth curve of 25 nm device becomes flat at higher voltage of -15 V and does not improve further by increasing the voltage from -15 V to -19 V. This means that device with 25 nm active layer thickness is limited by τ . The bandwidth does not enhance further by increasing the voltage. Increasing the active layer thickness reduces geometric capacitance leading to a decrease in τ . On the other hand, it also increases the charge transit time, thus there is a trade-off between transit time and τ as the active layer thickness is changed. Bandwidth improvement is less significant for the 100 nm device due to the less conductive thick active layer and an increase in transit time by increasing the active layer thickness. As a result, a 50 nm active layer was selected as an optimum thickness to obtain lower geometric capacitance compared to the 25 nm and smaller transit time than the device with the 100 nm active layer. Bandwidth as a function of active layer thickness is shown in Supplementary Fig. 3.

Figure 2b illustrates the responsivity as a function of voltage. At -2 V the device with a 50 nm active layer demonstrated the highest responsivity of 0.24 A/W followed by 25 nm and 100 nm with responsivity of 0.19 A/W and 0.18 A/W respectively. This higher responsivity of the 50 nm device can be attributed to the more optimized microcavity³¹ and a better charge balance³². The responsivity of each device structure was measured using a larger device (2 mm by 2 mm fabricated on glass substrate). This ensure that the active area exceeded the laser beam diameter, thereby enabling accurate optical excitation. Due to the lower thermal conductivity of the substrate, we measured over a smaller range of voltages.

Effect of electrode design: Trade-off between bandwidth and responsivity

Next, to investigate the effect of electrode design on bandwidth and responsivity, we compared the OPD with different electrodes (see Fig. 1). The frequency response and responsivity of the devices are influenced by the electrode design, including the thickness of the semi-transparent silver layer and the presence of interfacial layers.

Figure 3a illustrates the -6 dB bandwidth measured at -19 V as a function of silver thickness. A device with a 10 nm top silver electrode and 100 nm bottom reflective electrode (D1) exhibited a bandwidth of 79 MHz. Increasing the top silver thickness to 20 nm and the bottom reflective electrode thickness to 200 nm (D3) enhanced the bandwidth to 196 MHz. This means that the bandwidth of 50 nm D1 OPD is limited by the RC time too and increasing the thickness of the top and bottom electrodes leads to a reduction of τ and therefore bandwidth improvement. A further increase in the silver electrode thickness to 30 nm (D2) resulted in a further slight bandwidth improvement to 200 MHz. However, this device does not include the MoO₃/Au interlayer present in D1 and D3 and therefore represents a different electrode configuration. We note that the sheet resistance of thin silver electrodes was reported to reduce sharply when increasing the silver thickness from 7 nm to 20 nm, but this reduction of sheet resistance became less pronounced for the thicknesses beyond 20 nm^{29,33,34}.

Figure 3b presents the responsivity of OPDs with different electrode structures as a function of voltage. At -3 V device D1 with a silver thickness of 10 nm showed a responsivity of 0.27 A/W and this value was reduced to 0.12 A/W by increasing the silver thickness to 20 nm in device structure of

D3. Further increase in silver thickness to 30 nm in the device D2 led to a reduction in responsivity to 0.08 A/W. Device D4 with 20 nm silver thickness showed a responsivity of 0.14 A/W.

From Fig. 3a, b, it is evident that increasing silver thickness enhances device bandwidth but also introduces optical losses due to lower light transmission by the thicker silver electrode, leading to reduced responsivity. This establishes a trade-off between responsivity and bandwidth as shown in Fig. 3c. To balance these effects, a 20 nm silver thickness was selected as the optimal value. Bandwidth measurements of the device D3 at different voltages are provided in Supplementary Fig. 2. Further characterization of the D3 device (with an area of 0.05 mm²), including the microscopic image of the illuminated pixel, photocurrent and dark current measurements at -19 V, the stability test under continuous illumination at -19 V, the noise spectral density and detectivity analysis, as well as the capacitance and time-domain measurements, is presented in the Supplementary Information (Supplementary Fig. 6-11). Supplementary Fig. 8b shows that both the dark current and photocurrent gradually increase over time, while the modulation amplitude remains unchanged. This behaviour is likely caused by slow changes in the device during operation at a high reverse bias (-19 V), possibly due to heating. The drift affects the ON and OFF states similarly, so the modulation signal remains stable. Since these changes occur much more slowly than the communication signal, they do not affect the measured high-speed performance. However, such drift could reduce long-term signal stability in practical applications and should be considered in future device optimisation.

The stability test in Supplementary Fig. 10 shows that fluctuations during the first 100 minutes are likely due to the stabilisation under continuous illumination and high reverse bias, possibly because of heating and charge trapping effects.

It is important to note that the lower specific detectivity, D^* , in Supplementary Fig. 11 arises from the high reverse-bias operation required for high-bandwidth performance, which increases the dark current. D^* was extracted from long-duration noise measurements (650 s per segment) that capture low-frequency fluctuations so is not directly relevant to our high frequency communications measurements.

Effect of blocking layers

A comparison between D3 and D4 shows that adding 4,7-diphenyl-1,10-phenanthroline (BPhen) (as a hole blocking layer) and *N*, *N'*-di(naphthalene-1-yl)-*N*, *N'*-diphenylbenzidine (NPB) as (an electron blocking layer) to the structure results in a slight improvement in responsivity from (0.121 A/W to 0.144 A/W). However, this modification reduces -bandwidth (from 196 MHz to 89 MHz), attributed to an increase in transit time due to an accumulation of charge that reduces the internal electric field (Fig. 3a, b).

Effect of MoO₃/Au interlayer and Ag thickness

In our study, structures D1, D3, and D4 include MoO₃/Au beneath the semi-transparent Ag electrode. The main function of these layers is to act as a wetting layer to enable continuous ultra-thin Ag films. They do also slightly modify the optical cavity length, but much less than changes in silver thickness. Additional optical simulations were performed to verify this point and are provided in the Supplementary Information. Figure 3d shows that introducing the MoO₃/Au interlayer in the D1, D3, and D4 devices (which use thin Ag electrodes of 10–20 nm) produces a red-shift in the responsivity peak. Optical simulation in Supplementary Fig. 12 confirms that this shift

Fig. 2 | Device performance using different thicknesses of the active layer. **a** Bandwidth as a function of voltage using different active layer thicknesses and **b** Responsivity-Voltage characteristics for the devices measured with red laser light at 635 nm.

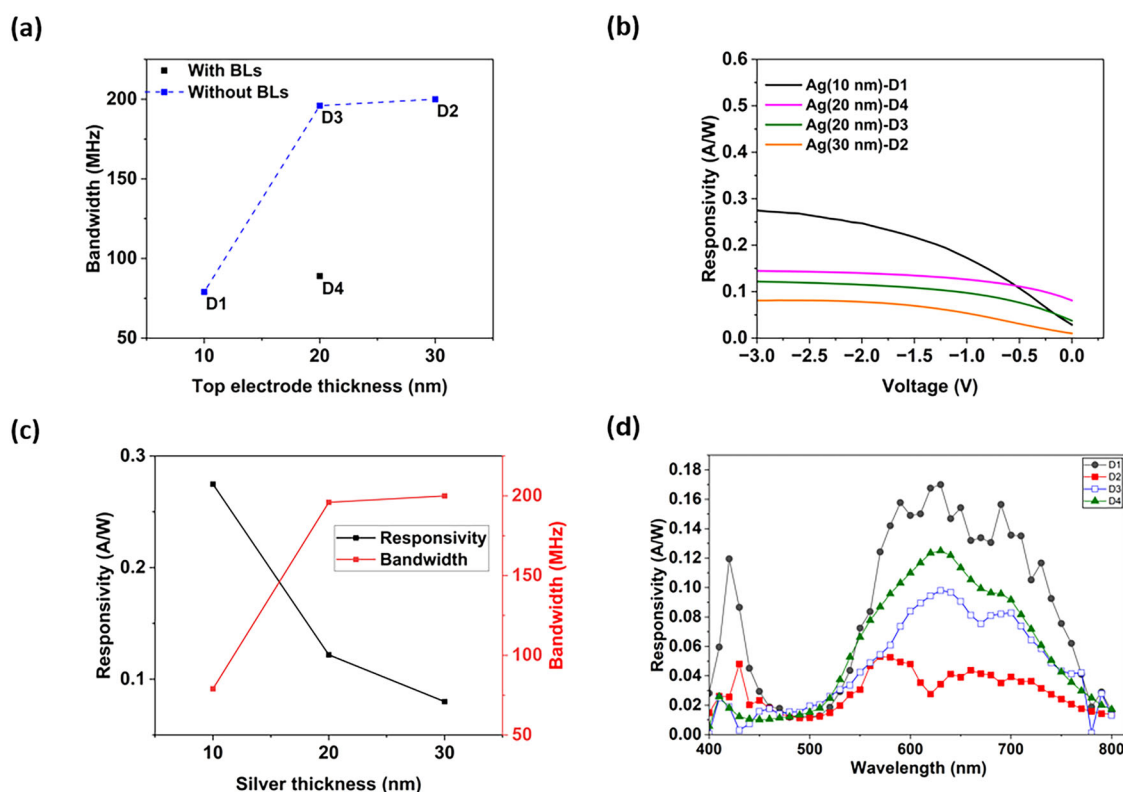
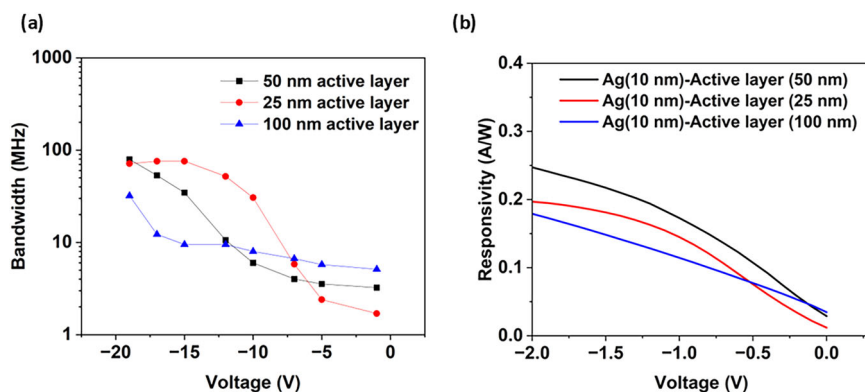


Fig. 3 | Effect of electrode design on device performance. **a** –6dB bandwidth at -19 V as a function of top electrode thickness and **b** responsivity as a function of voltage. **c** Responsivity and -6dB bandwidth as a function of silver (top electrode) thickness. **d** Spectral responsivity of OPDs. The active layer thickness for all devices is 50 nm.

Note: Device D2 (30 nm Ag) does not include the MoO₃/Au interlayer and, therefore, is not directly comparable to D1 and D3 for assessing the effect of silver thickness.

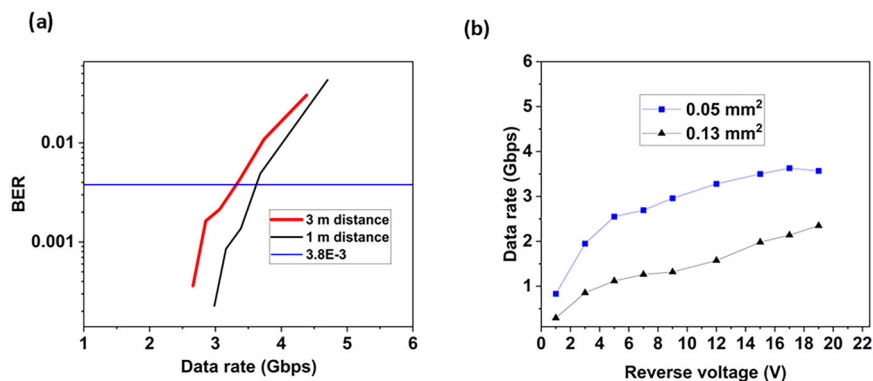
originates from a small increase in the optical path length and a slight change in the reflection at the metal/organic interface introduced by the MoO₃/Au layer. Experimentally, the 10 nm and 20 nm Ag devices exhibit nearly identical peak wavelengths, while the 30 nm Ag device (without MoO₃/Au) shows a blue shift. At the same time, the responsivity magnitude decreases clearly with increasing Ag thickness. As shown in Supplementary Fig. 12, changes in Ag thickness (from 10 to 30 nm) produce changes in reflectance, absorption and responsivity. These changes are much larger than those caused by MoO₃/Au at 30 nm silver thickness. Increasing Ag thickness reduces the spectral responsivity due to higher optical losses in Ag and increased reflectivity, which lowers the field inside the active layer. This behavior reflects the fact that although the MoO₃/Au interlayer can modify the cavity resonance even at 30 nm Ag, its effect becomes less visible in the measured responsivity because the increased Ag thickness strongly

attenuates incident light. Thus, at larger Ag thickness the responsivity is dominated by absorption losses in the metal. The optical behavior of the device is dominated by the silver thickness, while the MoO₃/Au interlayer has only minor optical influence.

Data rate measurements

As mentioned earlier in the paper, the D3 device with a size of 0.05 mm² and an active layer thickness of 50 nm demonstrated superior bandwidth performance compared to the other configurations. Therefore, this OPD structure was selected for data rate measurements. To test the OPD for data communication a red laser with a peak wavelength of 635 nm was used as the transmitter. Supplementary Fig. 4 illustrates the experimental setup of the OPD-based VLC system. The light illuminated by the red laser was collimated by a lens and then focused onto the OPD at the receiver side by

Fig. 4 | Communication performance of the D3 devices in a VLC link. **a** Data rates of the OPD with device structure of D3 and size of 0.05 mm^2 at FEC limit of 3.8×10^{-3} under 17 V reverse bias voltage in 1 m link and 3 m link. **b** Data rate as a function of reverse bias voltage for OPDs with active areas of 0.05 mm^2 and 0.13 mm^2 .



using another lens. The link distance was set to 1 m or 3 m. (Further details of data measurements are provided in the Method section). Figure 4a shows the bit error ratio as a function of data rate for the D3 OPD. For a link distance of 1 m, a data rate of 3.6 Gbps -17 V at a bit error ratio (BER) of 3.8×10^{-3} was obtained. For a 3 m data link we obtained 3.3 Gbps. In this work, a BER of 3.8×10^{-3} is considered the FEC threshold, with a 7% overhead (OH), where conventional coding can be used to ensure reliable, error-free data transmission.

Figure 4b presents the data rate as a function of voltage for OPDs with a size of 0.05 mm^2 and 0.13 mm^2 . At a reverse bias of 1 V and a 1 m link the data rates for the 0.05 mm^2 and 0.13 mm^2 devices were measured to be 833 Mbps and 294 Mbps, respectively. These values increased with voltage, reaching to maximum data rate of 3.6 Gbps and 2.3 Gbps at reverse bias of -17 V for the smaller and -19 V for the larger one. The signal-to-noise ratio (SNR) and bit-loading characteristics as a function of frequency for the D3 device with the size of 0.05 mm^2 are presented in Supplementary Fig. 5.

Data rate measurements for the different structures at -7 V and -19 V are summarized in Figs. 5a, 5b and Table 2. An OPD of size 0.05 mm^2 , based on the D2 structure with a 130 nm active layer thickness showed the lowest data rate of 0.64 Gbps and 1.67 Gbps at -7 V and -19 V, respectively.

In contrast, the D3 structure with the same size achieved the highest data rate of 2.66 Gbps and 3.55 Gbps at -7 V and -19 V respectively. Figure 5c compares the data rate of our D3 OPD to those reported in the literature^{6,18,21,35–40} as a function of bandwidth. Before this work, the highest data rate reported of OPDs was by Tavakkolnia et al.⁶ with a data rate of 147.5 Mbps for a single OPD and 363 Mbps using MIMO with four OPDs. Due to the bandwidth enhancement of our devices, data rates for a single OPD of 3.6 Gbps (over a 1 m link) and 3.3 Gbps (over a 3 m link) i.e. more than twenty times higher⁶ were achieved over a longer distance. Further details of this comparison are available in Supplementary Table 1 of the supporting information.

In contrast to conventional organic photodiode operation, which typically avoids strong reverse bias due to increased dark current and degradation effects, our devices are intentionally operated under extreme reverse bias conditions. This is enabled by the use of a silicon substrate acting as an efficient heat sink. Similar approaches to operating organic optoelectronic devices under non-standard conditions have been demonstrated in prior work, including high-speed OLEDs for optical communications¹³ and electrically driven OLED laser pumping⁴¹. Here, we exploit the trade-off between dark current and speed by prioritising temporal response, which is critical for communication applications. The increase in dark current is likely related to an increase in charge injection and device heating under a high reverse bias. These effects happen on long timescales so mainly affect the low-frequency response, rather than detection of our high-frequency communication signal. As a result, we achieve a record communication rate, demonstrating that operation under strong reverse bias can yield significant performance advantages despite the associated drawbacks. This behaviour distinguishes our approach from conventional organic photodiode design strategies.

Conclusion

In this work, we explored OPDs for VLC and investigated the factors affecting their bandwidth. We showed a trade-off between transit time and electrical time constant, τ , by adjusting the active layer thickness as well as a trade-off between bandwidth and responsivity by increasing the semi-transparent silver electrode thickness. By designing a structure to address these trade-offs and manage the speed limiting factors of the OPD, we achieved a record data rate of 3.6 Gbps over a 1 m link and 3.32 Gbps over a 3 m link for a single OPD receiver. This paper reports a breakthrough in the capability of OPDs as high-speed receivers for a wide range of VLC applications.

Methods

OPD fabrication

All layers were deposited by thermal evaporation using custom-made shadow masks in a vacuum chamber at a base pressure of 10^{-7} mbar (Angstrom Engineering, EVOVAC vacuum deposition system). A 200 nm silver layer was used as wiring electrodes and 200 nm aluminum⁴² layer was used as bottom electrodes, deposited on either glass or silicon substrates. A 60 nm electron transport layer of 4,7-diphenyl-1,10-phenanthroline (BPhen) was doped by cesium carbonate (CsCO_3). The active layer with different thicknesses consists of a blend including the zinc-phthalocyanine (ZnPc) as an electron donor and fullerene (C_{60}) as an electron acceptor with a ratio of 2:1. The hole transport layer was a 210 nm layer of 2,2',7,7'-tetra-(perfluoronaphthalene-2,6-diylidene) dimalononitrile ($\text{F}_6\text{-TCNNQ}$) (4 wt %). A 7 nm thick MoO_3 and a 1 nm gold followed by silver with different thicknesses were used as top semi-transparent electrodes.

A 70 nm thick layer of *N*, *N'*-di(naphthalene-1-yl)-*N*, *N'*-diphenylbenzidine (NPB) was used as the capping layer. For a device with blocking layers, a 10 nm thick BPhen and a 10 nm thick NPB layer were used as hole and electron blocking layers respectively. After evaporation, all OPDs were encapsulated in a nitrogen glovebox using glass lids, epoxy glue (Norland Optical Adhesive 68) pre-baked under nitrogen conditions, and a moisture getter (Dynic Corporation, HD-071210T-50S). To dissipate Joule heating at high reverse voltages the OPDs were mounted on a custom-made holder equipped with a heat sink and a fan.

OPD characterization

The photodiode responsivity was tested by using a red laser and a source measure unit (Keithley 2400). Spectral responsivity was measured at zero bias by illuminating the photodiodes with light from a xenon lamp (Bentham, Model 605) in combination with a monochromator (Bentham TMC300).

The frequency response of OPDs was measured using a red laser with a wavelength of 635 nm (ROHM, RLD63NPC8) at an optical power of 0.2 mW. A pair of lenses was used to collimate the light from the laser and focus down to the OPDs. A vector network analyser (Keysight, E5061B) was used to measure the frequency response. The laser was directly connected to

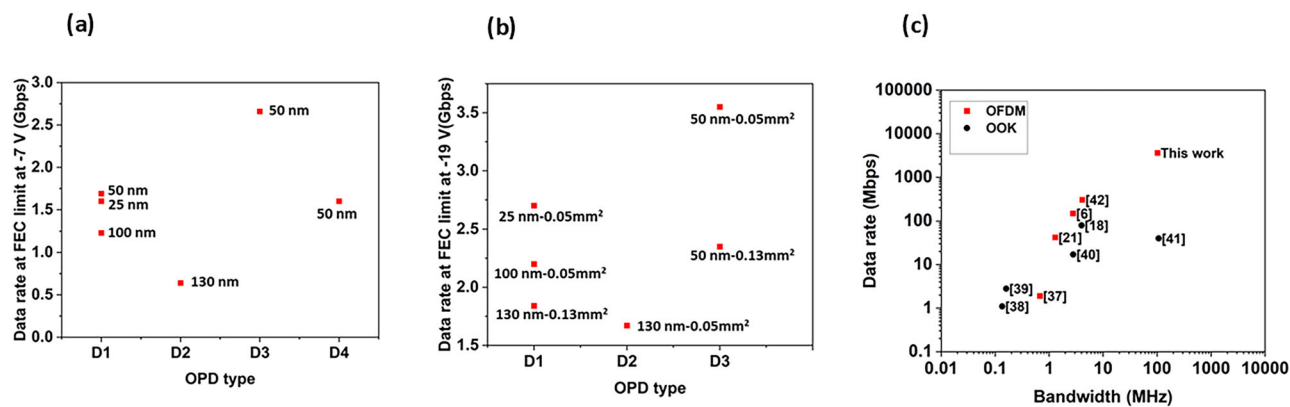


Fig. 5 | Summary of data rate measurements for different types of OPDs. a At -7 V and **b** at -19 V. **c** Data rate comparison of our OPD using the D3 structure with the literature devices as a function of bandwidth.

Table 2 | Summary of bandwidth and data rate measurements for 0.05 mm² devices using different thicknesses of active layer and electrodes at -19 V

Device type	Active layer thickness (nm)	Top electrode	Bottom electrode (nm)	Bandwidth (MHz)	Data rate (Gbps)
D1	25	MoO ₃ -Au-Ag (10 nm)	Al (100)	71	2.7
D1	100	MoO ₃ -Au-Ag (10 nm)	Al (100)	32	2.2
D2	130	Ag (30 nm)	Al (200)	34	1.67
D3	50	MoO ₃ -Au-Ag (20 nm)	Al (200)	196	3.55

port 1 of the network analyzer. To measure the frequency response at different bias voltages while blocking the DC signal, a bias tee (Mini-Circuit, ZFBT-4R2GW +) was employed. The RF port was connected to port 2 of the network analyzer, which was internally terminated with 50 Ω . The DC port was connected to the source measure unit. The combined port was connected to the OPDs. The signal transmission coefficients known as “S21” were measured at different frequencies. The bandwidth for OPDs was estimated as the frequency at which the electrical power decreases 6 dB from the value at low frequency. To ensure reproducibility, bandwidth measurements were conducted on three individual pixels for each device structure, and the corresponding data are provided in the Supporting Information (Supplementary Fig. 13).

Responsivity Measurements of 0.05 mm² device at -19 V:

Responsivity measurements were performed on a device with the D3 architecture. To ensure accurate responsivity measurements, the illuminated region was verified using a custom-built optical microscope to monitor the laser alignment during measurements. Microscopic imaging confirmed that the laser spot was fully confined within the photodiode's active region and that the spot diameter remained smaller than the defined active area, preventing beam over-fill and avoiding artificially inflated responsivity values, as emphasized in the consensus guidelines⁴³.

Prior to illumination, device was biased in reverse and allowed to stabilize for approximately 15 minutes to ensure that the dark current reached a quasi-steady state, following recommended procedures that highlight the need to allow for dark-current stabilization before any photocurrent extraction.

Responsivity was measured using a laser diode (RLD63NPC8) delivering 0.2 mW of optical power at the device surface. Given the device active area of 0.05 mm² (125 μ m by 400 μ m), the corresponding optical intensity averaged over the area of the device was 4×10^3 W/m². The photodiode was biased using a Keithley 2401 Source Meter, which provided a stable reverse-bias voltage during measurement. To enable reliable separation of photocurrent from dark current, the incident laser was modulated by a motor-driven mechanical shutter, generating periodic on/off illumination. This

approach follows the consensus recommendation to use controlled optical modulation to enable accurate extraction of photocurrent amplitude.

Response time and capacitance measurements:

Capacitance in our study was from the imaginary part of the small-signal admittance(B), measured using a vector network analyser (Keysight, E5061B) with a single port reflection mode.

The temporal response of the photodiode was evaluated using optical pulses generated by driving the laser diode with an electrical pulse generator (Agilent B11110A). The laser was driven using 50 ns electrical pulse, resulting in corresponding 50 ns optical excitation. The start and end points used for rise-time and decay-time extraction are indicated by arrows. The initial overshoot was excluded as it arises from parasitic and measurement effects. The decay time was measured between the appropriate signal levels representing the intrinsic photodiode response. The measurement was performed using a high bandwidth 500 MHz oscilloscope (Tektronix TDS 3052 C).

Lifetime test:

The long-term operational stability of the photodiodes was assessed using a continuous-bias lifetime test. The device was held under a constant reverse bias of -19 V and continuous illumination was applied, after which the photocurrent was recorded for 200 minutes. The bias was supplied and the current was continuously monitored using a Keithley 2401 source meter. At first, the current exhibits an initial decrease, which is attributed to rapid photo-induced charge redistribution or early detrapping processes occurring within the device. Following this brief dip, the current increases gradually, consistent with trap filling and interface conditioning under sustained reverse bias. After reaching a maximum, the current slowly decreases and approaches a stable value, representing the intrinsic long-term behavior of the photodiode under continuous electrical and optical stress.

Experimental setup for communication measurements in OPD-based VLC system

The performance of the OPD-based VLC system was evaluated over link distances of 1 m and 3 m. The data transmission utilized direct-current-

biased optical orthogonal frequency-division multiplexing (DCO-OFDM) with quadrature amplitude modulation (QAM), generated in MATLAB and sent to an arbitrary waveform generator (AWG, Keysight M8195A). To maximize the achievable data rate, an adaptive bit and power-loading algorithm based on the Hughes-Hartogs method was applied to each subcarrier. This algorithm optimized system performance and spectral efficiency by dynamically adjusting the modulation scheme for individual subcarriers. Higher-order modulation was assigned to subcarriers with high signal-to-noise ratio (SNR), while lower-order modulation was applied to those with low SNR. The transmitter used channel state information to estimate the SNR of each subcarrier and select the optimal bit allocation strategy accordingly. The modulated signal, combined with a DC current from a power supply (Keysight 36312 A), was fed into a bias tee (Mini-Circuits, ZFBT-4R2GW +). The output signal from the bias tee was applied to a red laser (ROHM, RLD63NPC8) with a central wavelength of 635 nm with an output power of 10 mW. A DC bias of 60 mA at 2.5 V was used to operate the laser. To ensure linear operation and maximize the average SNR across the modulation bandwidth, the peak-to-peak voltage of the OFDM waveform was set to 1 V on the AWG. A 75 mm aspheric condenser lens (Thorlabs, ACL7560U-A) collimated the emitted light from the laser, and on the receiver side, a similar lens on the receiver side focused the collected light onto the OPDs. On the receiver side, the output signal from the OPD and the reverse voltage were combined in a bias tee, and the output alternating current (AC) signal was amplified using a low noise amplifier (ZFL-1000LN +). The amplified signal was captured by an oscilloscope (Keysight, UXR0104B) and transferred to a laptop, where the data were processed using MATLAB software for further analysis.

SNR and bit loading

The SNR and adaptive bit-loading results shown in Supplementary Fig. 5 were obtained for transmission distances of 1 m and 3 m at a reverse-bias voltage of -17 V at a BER threshold of 3.8×10^{-3} . Adaptive bit allocation was performed using the Hughes-Hartog (HH) algorithm, which distributes the available transmission power among the OFDM subcarriers according to their measured SNRs⁴⁴. For each subcarrier, the algorithm determines the incremental power required to increase the modulation order by one bit while maintaining the specified BER constraint. Starting from the minimum modulation level, the HH algorithm iteratively assigns an additional bit to the subcarrier requiring the lowest incremental power. The allocation process continues as long as the subcarrier SNR remains sufficient to support a valid modulation level under the FEC threshold; in the present experiment, subcarriers with SNR values approaching approximately 0 dB represent the lower limit for useful bit loading. Consequently, subcarriers with higher SNR are assigned higher-order modulation, whereas subcarriers experiencing poorer channel conditions are allocated fewer bits or are left unloaded.

In the experimental results, the low-frequency region exhibits the highest SNR and therefore supports modulation orders of up to 9 bits per subcarrier. As the frequency increases, the progressive reduction in SNR, owing to the channel characteristics. The staircase-like bit-loading profiles in Supplementary Fig. 5 thus directly reflect the frequency-selective SNR response of the OPD receiver. For both the 1 m and 3 m links, the low- and mid-frequency subcarriers maintain sufficiently high SNR to support relatively high-order modulation, whereas the pronounced SNR degradation toward the upper end of the bandwidth progressively restricts the corresponding subcarriers to lower modulation orders and, where the SNR becomes insufficient, causes them to be unloaded.

This adaptive allocation enables more efficient utilisation of the available bandwidth and transmission power than a fixed-modulation scheme because the available spectral capacity is preferentially assigned to subcarriers with favourable channel conditions. Consequently, the achievable OFDM data rate is determined by the sum of the bits allocated across all active subcarriers, after accounting for the associated OFDM, FEC, and other system overheads.

Data availability

The research data underpinning this publication can be accessed at: <https://doi.org/10.17630/01a28039-e9b2-4907-bb5d-faf7954572bf>.

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Author contributions

I.D.W.S., H.H. and G.A.T. conceived of the project and supervised the research. M.R. fabricated and developed OPDs discussing with K.Y. B.M. performed the data rate measurements of OPDs. K.Y. packaged up the laser diode used for the bandwidth measurements and the transmitter of the data rate measurement.

Competing interests

H.H. is a founder, director and shareholder of pure LiFi, a company developing LiFi technology.

Additional information

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